A Cosine Transform and Minority Graph-based Network (CTMGN) framework is introduced for multiclass imbalanced image classification. The method combines DCT-based dimensionality reduction, a graph constructed from minority-class prototypes, a learnable adjacency matrix, cross-attention, and a two-stage classification head. The evaluations are on the approach of imbalanced variants of SVHN and CIFAR-10, provide component-level ablation experiments, compare computational requirements, and apply McNemar's test. The emphasis on modelling relationships among minority classes rather than relying only on resampling or loss reweighting is interesting. The lightweight architecture and reported reductions in parameter count and inference time are also potentially valuable.

Major issues -

- The proposed method is compared mainly with conventional CNN, attention-CNN, and GCN implementations. However, these are not representative of current specialised imbalance-learning approaches. Some manageable number of recognised baselines, such as focal loss, class-balanced loss, LDAM with deferred reweighting, Balanced Softmax or logit adjustment must be included.
- Different learning rates, batch sizes and training configurations are reported across the ablation and comparison experiments. Furthermore, the competing architectures differ greatly in parameter count, making it difficult to determine whether improvements arise from the proposed mechanism or implementation choices. These differences were particularly seen in section 3.2 and 3.3 in SVHN and CIFAR datasets.
- Both training and test sets appear to be artificially imbalanced by discarding samples. Altering the standard test distribution can make results less comparable with previous studies and may cause overall accuracy to depend strongly on the selected test proportions. Alternatively, report results on both the complete test set and the imbalanced test set, with exact class-wise sample counts.
- For SVHN, CTMGN improves overall accuracy from 78.50% for MGN to 78.59%, while average precision slightly decreases. For CIFAR-10, the improvement over MGN is also modest. Results from a single run do not demonstrate that these gains are stable. Repeat the main experiments with at least three independent random seeds and report mean ± standard deviation.
- It has been stated that low-frequency coefficients are retained, but the exact coefficient-selection pattern and compression ratio are not clearly reported. Conversion from RGB to grayscale may also remove useful colour information, particularly for CIFAR-10. Add a concise sensitivity analysis for two or three compression levels and discuss the effect of grayscale conversion.
- CTMGN receives compressed one-dimensional features, whereas CNN and attention-CNN baselines process full RGB images. Consequently, the reported runtime advantage is partly caused by different input representations. Compare runtime with a baseline that also receives the same DCT features, and report parameter count, FLOPs or MACs, peak memory and average inference time per image under identical hardware and batch settings.

Minor Issues -

- Table 4 and Table 6 have appear to report only the McNemar Z-statistics rather than complete test results. Report p-values, the significance threshold and any correction used for multiple pairwise comparisons.
- Several methodological descriptions are difficult to follow because tensor dimensions are not consistently provided. Add the dimensions beside the main variables and network layers.
- Explain why minority-class Gaussian-noise augmentation is used only for SVHN. Either apply equivalent augmentation consistently or provide results showing that it does not confound the comparison.